

Design Document: Introduction to Game Programming

Working-Title: Thrust-Clone

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Feature 1: Anti-Aircraft Gun

1. What the feature does and what it brings to the game

An anti-aircraft gun, a turret, is a stationary enemy unit embedded within the level and designed to introduce an active environmental threat to the player. It detects the rocket upon entering a predefined range and fires projectiles at it once aligned. It requires the player to navigate the rocket carefully around the areas or destroy the turret to proceed collecting objects and landing safely. This feature aims to bring the game a challenge. Figure 1 shows some visual references can be used as inspiration for the turret's design.



Figure 1: Turret Asset Examples

2. Brief Architecture

The turret is placed at a fixed position in the level; it does not move, only rotates in place to aim at the rocket. It is made up of a sprite for the visual part and a detection area that checks whether the rocket has entered its range.

In `_PhysicsProcess`, the turret checks if the rocket is inside this detection area. If it is, the turret rotates towards the rocket's current position every physics frame, so the aim keeps updating as the rocket moves. Once the turret is properly aligned with the rocket, it spawns a projectile at fixed time intervals by using a timer variable.

The turret and the projectile are implemented as two separate scenes, since a projectile needs to be instantiated fresh every time the turret fires, and removed again once it hits something or leaves the screen. The turret holds a reference to the projectile scene and, when it is time to shoot, creates a new instance of it, places it at the turret's position, and gives it the direction it should move in. From that point on, the projectile moves and handles its own collision detection independently of the turret. When it hits the rocket, it calls into the Combat & Damage System (Feature 2), which is responsible for reducing the rocket's health, and the projectile then removes itself from the scene.

3. Dependencies and Risks

This feature depends on the Combat & Damage System to resolve the outcome of a successful hit, so that the projectile collisions have a gameplay effect.

A further dependency is the correct configuration of collision layers and masks to ensure projectiles interact only with the rocket.

Additionally, the exact number of turrets to be placed in each level has not been finalized yet and may need adjusting depending on the overall difficulty balance of the game. Since turret placement is tied to level design, poorly placed turrets could make certain areas feel unfair or impossible to pass without taking damage, so their positions will also need to be tested and adjusted alongside the levels themselves.

4. Acceptance Criteria

- A turret remains inactive while the rocket is outside its detection range
- A turret stays fixed in its position; it does not move, only rotates to aim at the rocket.
- A turret fires projectiles at a fixed interval once correctly aligned with the rocket.
- Projectiles move along their intended trajectory and are removed upon impact or when leaving the visible play area.
- A projectile hitting the player correctly triggers the Combat & Damage System (Feature 2)